

Supplementary Material

Structurally-Calibrated Functional Attribution for Audit Prioritization in Knowledge-Intensive Reasoning

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This Word file provides a readable supplementary overview for editorial and reviewer navigation. The complete mathematical notation, equations, algorithms, tables, figures, and exact captions remain in supplementary.tex inside latex_source.zip, which is the authoritative supplementary source for reproduction.

The supplement supports the DKE positioning of SC-FMA as a knowledge-engineering layer that converts structured knowledge-intensive traces into graph-aware audit records for fixed-budget knowledge curation. It does not add validation stages beyond the manuscript evidence package.

Supplement Scope

- Extended SCU derivations and optimization diagnostics, including KKT conditions, variance-reduction reasoning, bottleneck behavior, and redundancy-driven monotonicity cases.
- Additional controlled-stage audit-priority consistency, ablation, sensitivity, efficiency, and proxy-label independence results.
- PRM800K variant details and fixed-budget audit-prioritization readout for annotation-signal evidence.
- Knowledge-graph and constructed-label diagnostics, including Countries-KG, WebQSP fixed-schema trace audit, and MuSiQue feasibility material.
- Reproducibility notes covering configuration, seeds, scripts, locked artifacts, and interpretation limits.

Content Map

Supplement Section	What It Contains	Role in the Paper
Extended Proofs and Derivations	Documents the SCU objective behavior, bottleneck protection, variance-reduction intuition, and redundancy effects.	Supports method transparency without changing the main mathematical definitions.
Additional Validation Results and Figures	Reports controlled synthetic audit-priority consistency by trace size, component contributions, sensitivity grids, and scaling diagnostics.	Places synthetic evidence as controlled calibration rather than the primary real-data claim.
Main-Text Material Moved to Supplementary	Keeps notation, algorithms, hyperparameters, and efficiency details available outside the compressed manuscript.	Preserves reproducibility while keeping the main text focused on DKE framing.

Supplement Section	What It Contains	Role in the Paper
PRM800K Variant Details	Summarizes Ridge, QP, and projection variants using w_struct as the locked fidelity input.	Supports fixed-budget audit prioritization under annotation signals.
Knowledge-Graph and Trace Diagnostics	Records Countries-KG, WebQSP, MuSiQue, and supplementary figure diagnostics.	Separates representation-backend and feasibility diagnostics from external deployment claims.
Implementation and Reproducibility	Lists reproduction steps, convergence diagnostics, graph parameter sensitivity, and proxy-label independence checks.	Supports auditability of the evidence package and artifact review.

Key Evidence Carried by the Supplement

- PRM800K locked split: 4,417 samples and 34,219 labeled steps are used for the real-data annotation-signal audit-prioritization stage.
- Rule-derived audit-target retrieval reports fixed-budget audit-target coverage improvement from 0.235 to 0.699 over the scalar-only view.
- Countries-KG evaluates knowledge-graph representation sensitivity by comparing structural-dependency allocation across graph backends.
- Controlled synthetic results are used for calibration, ablation, sensitivity, and error analysis rather than as the opening real-data evidence.

Scope Notes

The supplementary evidence is interpreted consistently with the main manuscript: SC-FMA produces graph-aware audit records for prioritizing review under limited curation budgets. Deployment validation, downstream model-training gains, external-transfer claims, and causal identification are outside the reported evidence package.

Process Reward Models and process-supervision datasets appear only as sources of step-level annotation signals. The paper contribution remains the knowledge-audit representation and structural calibration workflow.

Reproducibility Inventory

- The source bundle contains manuscript.tex, supplementary.tex, references.bib, CAS template files, and all referenced figures.
- The controlled synthetic stage uses five fixed seeds and archived hyperparameter grids; exact values and tables are preserved in supplementary.tex.
- Variant selection and interpretation follow the locked PRM800K v3.6/v3.8 evidence status and do not rely on unreported downstream training.

- The source package and verifier enforce the DKE transfer boundary and keep obsolete prior-venue wording out of the upload package.

Source of Record

For mathematical symbols, equations, algorithms, table values, and figure captions, use `final_source/supplementary.tex` in `latex_source.zip` as the source of record. This Word file is intentionally optimized for readable submission-package review and consistency checking.

Data & Knowledge Engineering supplementary overview